

Robust Markowitz Portfolio Optimization in a Broad ETF Universe

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Abstract—A large aspect of finance is asset management and allocation, with considerations including the balance between risk and return, diversity, and stability. This is precisely the matter that portfolio construction seeks to deal with. Our project seeks to formulate a robust Markowitz framework that is more stable and effective than the classical Markowitz framework. Classical Markowitz operates as a mean-variance framework that balances expected return and variance, but is very sensitive to estimation error given that it relies on indirect analysis of historical data. The analysis in this project is based on a broad multi-asset exchange-traded fund (ETF) universe, for which we evaluate data from 2015 to 2025 using a two-year rolling training window. The setup for the robust framework included full investment and maximum weight caps.

Our robust Markowitz implementation had some advantages. These include noticeably reduced volatility and lower maximum drawdown relative to the classical Markowitz model. However, neither the robust nor the classical Markowitz strategy produced better overall results than the equal-weight strategy, which is a simple diversified benchmark.

Index Terms—Portfolio optimization, Markowitz optimization, robust optimization, exchange-traded funds, backtesting.

I. INTRODUCTION

Portfolio construction is a delicate operation that those in financial advisory and wealth management roles conduct every day. It requires investors to choose the most attractive investments that produce the highest returns while controlling trade volume, limiting volatility, and remaining constrained by a finite amount of capital.

The Markowitz mean-variance framework is an elementary approach that is fairly well known. Instead of evaluating assets one at a time, the model takes in expected returns and a covariance matrix of asset returns to balance the returns against risk. Despite the benefits that this approach brings by turning diversification into a mathematical decision, it also comes with significant downfalls. For one, the optimizer is heavily dependent on future estimates that are extrapolated from past data, which is heavily subject to noisiness. This is particularly problematic in a broad ETF universe, where the diversity in asset types can make the data especially noisy.

Originally, as detailed in our project proposal, we sought to determine whether robust portfolio construction could be made more effective through deliberate universe selection. We proposed correlation-based clustering as a possible step during asset selection to remove redundant ETFs. Our final project simplifies this objective by cutting out the step of optimizing universe selection and instead focuses on building a backtesting pipeline that compares an equal-weight framework, a classical Markowitz framework, and a robust Markowitz

framework. In terms of technicals, we used `cvxpy` instead of `cvxportfolio`, which meant that our implementation lacked some portfolio-specific modeling features but gave us direct control over the optimization problems.

The research question for the final project asks whether a robust Markowitz framework is able to perform better than both a classical Markowitz framework and an equal-weight framework in a broad ETF universe. We found that robust Markowitz produced reduced volatility and a lower maximum drawdown relative to classical Markowitz, but both approaches were trumped by the equal-weight approach. This result shows that more sophisticated optimization techniques are not automatically more effective than simple techniques.

II. BACKGROUND AND RELATED WORK

Portfolio construction is a problem centered around deciding how much capital one should allocate to each given asset in an investment universe. A portfolio is typically represented by a vector of weights, w , wherein each entry w_i represents the fraction of the total capital invested in asset i . The main challenge of portfolio construction is that investors want high returns, but simultaneously want to limit risk by avoiding overconcentration and portfolios that change too aggressively over time.

Markowitz’s portfolio optimization model was one of the first major attempts to convert portfolio selection into a mathematical optimization problem [8]. Rather than considering each asset one by one, Markowitz examined the portfolio as a whole and subsequently had the insight that risk depends not purely on the volatility of each individual asset; it also depends on how the growth of those assets is correlated. This insight laid the foundation for mean-variance optimization and heavily influenced modern portfolio theory.

In Markowitz optimization, the investor chooses weights w with the goal of maximizing expected return while penalizing portfolio variance:

$$\max_w \quad \mu^T w - \gamma w^T \Sigma w \quad (1)$$

subject to

$$\mathbf{1}^T w = 1, \quad w \geq 0. \quad (2)$$

In this expression, μ is the vector of expected asset returns, Σ is the covariance matrix of asset returns, and γ is a risk-aversion parameter. Hence, $\mu^T w$ represents the expected return of the portfolio, while $w^T \Sigma w$ represents the variance of the portfolio. The constraint $\mathbf{1}^T w = 1$ forces the portfolio to be fully invested. The constraint $w \geq 0$ is a long-only constraint, which means that the investor can only buy and hold assets.

The utility of this model lies in the fact that it allows us to treat diversification as a formal optimization problem. However, its main weakness is that the quantities it relies on are not known in advance. The expected return vector μ and the covariance matrix Σ must be estimated from historical data, and these estimates can be inaccurate because they are based on noisy data. If they are inaccurate, then the optimizer may cause the model to overfit the training data, meaning that the portfolio performs well in-sample but poorly out-of-sample. Boyd et al. describe this as one of the central practical issues in modern Markowitz portfolio construction [2].

This issue is particularly relevant in a broad ETF universe. Many funds have overlapping exposures. ETFs that track similar indexes or sectors may be highly correlated, and differences in estimated returns or covariances between them may be due to noise rather than any meaningful economic difference between the ETFs themselves. A Markowitz optimizer can amplify these errors by shifting weight between similar ETFs. This creates weight instability and high turnover, which makes the portfolio less practical once transaction costs and barriers to implementation are considered.

This motivates robust optimization, which attempts to address this problem by making the portfolio less sensitive to estimation error. It does so by adding an additional penalty term to discourage the portfolio from being fragile. A general robust Markowitz objective can be written as

$$\max_w \quad \hat{\mu}^T w - \gamma w^T \hat{\Sigma} w - \lambda R(w), \quad (3)$$

subject to

$$\mathbf{1}^T w = 1, \quad 0 \leq w_i \leq w_{\max}. \quad (4)$$

Here, $\hat{\mu}$ and $\hat{\Sigma}$ represent the estimated return vector and covariance matrix. $R(w)$ is a robustness penalty, weighted by λ . Setting the upper bound of w_i to $w_{\max}$ prevents the optimizer from placing too much weight on a single ETF. In essence, the robust model is more cautious than the classical model and avoids portfolios that depend too heavily on uncertain estimates. This framing is closely related to Goldfarb and Iyengar's robust portfolio selection work, which treats robust optimization as a response to modeling and statistical errors [6].

Our project fits into this background by testing whether this theoretical advantage causes better empirical performance. We compare equal weight, classical Markowitz, and robust Markowitz strategies using historical ETF data and a rolling backtest to evaluate each strategy's forecasting ability. At each quarterly rebalance, the optimized strategies estimate returns and covariances from past ETF returns, compute new weights for the portfolio, and then use those weights until the next quarterly rebalance. This allows us to evaluate whether robust Markowitz optimization improves return, volatility, Sharpe ratio, maximum drawdown, and turnover when compared with both classical Markowitz and equal weight.

III. METHODOLOGY

Our project is based on a broad set of approximately 150 ETFs that cover major asset classes including domestic and international equities, bonds, real estate, commodities, and

dividend or factor ETFs. We built a complete backtesting pipeline in Python that downloaded adjusted ETF price data using `yfinance`, cleaned it, converted the prices into daily returns, and ran quarterly rebalanced strategies. The price data runs from 2015 to 2025. Because the optimized strategies require a two-year rolling training window, the out-of-sample backtest begins after enough historical data is available and is rebalanced every quarter. The implementation made use of a 504-trading-day rolling estimation window, as well as a 10% maximum ETF weight cap, and quarterly rebalancing. At each rebalance date, the model estimates expected returns and covariance, solves for portfolio weights, and holds the portfolio until the next date, at which point the process repeats. In the final version of the project, the covariance estimates for the optimized strategies were based on iterated exponentially weighted moving average forecasts rather than plain rolling sample covariance [3].

The project compares three optimization strategies. The equal-weight strategy assigns the same weight to every ETF in the universe. This strategy does not attempt to estimate which ETFs will outperform; it simply provides a diversified benchmark.

The classical Markowitz strategy solves the following mean-variance optimization problem over the ETF universe:

$$\max_w \quad \hat{\mu}^T w - \gamma w^T \hat{\Sigma} w \quad (5)$$

subject to

$$\mathbf{1}^T w = 1, \quad 0 \leq w_i \leq w_{\max}. \quad (6)$$

Here, $\hat{\mu}$ is the estimated return vector from the rolling training window, $\hat{\Sigma}$ is an IEWMA covariance estimate from the same window, γ controls risk aversion, and $w_{\max}$ is the maximum weight cap for each ETF. The constraints require the portfolio to be fully invested, long-only, and not overly concentrated in a single ETF.

The robust Markowitz strategy makes the classical approach more conservative by adjusting both the expected return estimate and the covariance estimate. In our implementation, this is represented as

$$\max_w \quad (\hat{\mu} - \rho s)^T w - \gamma w^T \hat{\Sigma}_{\text{rob}} w, \quad (7)$$

subject to the same full-investment, long-only, and maximum-weight constraints. Here, s is the vector of estimated return standard errors, ρ controls the strength of the return haircut, and

$$\hat{\Sigma}_{\text{rob}} = \hat{\Sigma} + \kappa \text{diag}(\text{diag}(\hat{\Sigma})) \quad (8)$$

adds a diagonal covariance uncertainty adjustment. In the final version of the project, $\hat{\Sigma}$ for the robust strategy is based on a combined multiple IEWMA covariance forecast. This makes the robust strategy more cautious than the classical version by penalizing assets with more uncertain return estimates and by inflating risk along the diagonal of the covariance matrix.

Each strategy is individually evaluated using the following metrics: cumulative return, annualized return, annualized volatility, Sharpe ratio, maximum drawdown, average quarterly turnover, and final portfolio value. The performances are then compared to each other, from which we draw conclusions about their effectiveness relative to each other.

IV. RESULTS

The main result is that the equal-weight strategy performed best overall. It achieved the highest cumulative return, highest annualized return, highest Sharpe ratio, and lowest turnover. Classical Markowitz and robust Markowitz both ran successfully after fixing numerical issues in the optimizer, but both underperformed equal weight in this setup.

Fig. 1 shows the clearest difference between the strategies. Equal weight grew from an initial value of 1.00 to a final portfolio value of 2.1637, corresponding to a cumulative return of 1.1637. Classical Markowitz finished at 1.3795, and robust Markowitz finished at 1.1478. In other words, equal weight more than doubled the initial portfolio value, while the two optimized strategies produced much smaller gains.

This result suggests that the optimized strategies did not keep pace with the equal-weight portfolio in this implementation. Classical Markowitz remained sensitive to changing estimates and produced only modest gains, while robust Markowitz was more conservative and captured even less of the upside from the market recovery and growth periods after 2020.

Fig. 2 adds nuance to the portfolio value results. Classical Markowitz had the worst maximum drawdown at -32.27% . Equal weight had a maximum drawdown of -30.66% , while robust Markowitz had the best maximum drawdown at -24.80% . This means the robust strategy did provide meaningful downside protection, even though the classical strategy did not.

However, this lower drawdown came at the cost of much lower return. The robust Markowitz strategy spent long periods below its prior peaks, especially around the 2022–2024 period, and did not recover nearly as strongly as equal weight. Classical Markowitz also underperformed equal weight and experienced deeper drawdowns despite producing much smaller gains. This shows that maximum drawdown alone does not capture the full investment experience. A strategy can have a smaller worst drawdown but still perform worse overall if it fails to participate in recoveries.

Fig. 3 shows that turnover was one of the most important differences between the strategies. Equal weight had an average quarterly turnover of only 0.0244. Classical Markowitz had an average turnover of 0.4612, and robust Markowitz had an average turnover of 0.5190. This means the optimized strategies changed their portfolios far more aggressively at each rebalance.

This high turnover is a practical weakness. Even though transaction costs were not the central focus of this implementation, a strategy that changes close to one-half of the portfolio each quarter would be significantly more expensive to trade. In the final results, robust Markowitz actually had higher turnover than classical Markowitz, which further bolsters the conclusion that equal weight was far more stable and practical from a trading perspective.

The metrics in Table I confirm the visual results. Equal weight produced an annualized return of 9.25%, compared with 3.76% for classical Markowitz and 1.59% for robust Markowitz. Equal weight also had the highest Sharpe ratio

at 0.6594, compared with 0.2276 for classical Markowitz and 0.1626 for robust Markowitz. Even though robust Markowitz had the lowest annualized volatility, its return was too low to compensate for that reduction in risk.

Robust Markowitz improved risk control relative to classical Markowitz. It reduced annualized volatility from 16.50% to 9.79% and improved maximum drawdown from -32.27% to -24.80% . However, it also reduced annualized return from 3.76% to 1.59% and reduced Sharpe ratio from 0.2276 to 0.1626. Therefore, in this version of the project, robust optimization improved downside metrics relative to classical Markowitz but did not improve overall risk-adjusted performance.

Fig. 4 shows how the optimizer allocated across ETFs over time. The darker blocks indicate periods when certain ETFs received larger weights, often near the maximum weight cap. The heatmap shows that the robust strategy did not simply hold a smooth diversified portfolio. Instead, it shifted between groups of ETFs over time. This helps explain the high turnover result: even with a more conservative covariance adjustment, the optimizer still changed its preferred assets as the rolling training window updated.

The heatmap also shows why further work on universe selection could matter. If many ETFs are correlated or overlapping, the optimizer may rotate between similar funds as their recent estimated returns change. A clustering step could potentially reduce this behavior by selecting representative ETFs before optimization.

V. DISCUSSION

This report has repeatedly placed emphasis on the takeaway that a more sophisticated optimization method like robust Markowitz does not outperform a simple equal-weight framework by default. Indeed, the equal-weight framework avoided problems with estimation error, kept turnover low, and captured broad market growth. However, these findings should not be conflated with the idea that Markowitz optimization is entirely ineffective. In the final results, the robust framework reduced volatility and maximum drawdown relative to both equal weight and classical Markowitz, even though it still underperformed on return and Sharpe ratio.

These improvements make the robust portfolio less vulnerable to extreme outcomes and reduce the likelihood of issues related to historical overfitting. However, this can also be interpreted as a shift along the spectrum from aggressive to conservative. The robust technique makes the portfolio more stable, but that may also be what causes it to miss the riskier upside that would eventually bring positive performance.

This result is also reasonable, given the fact that the universe consisted entirely of ETFs, which are already diversified baskets of assets. In a riskier universe of individual stocks or more volatile instruments, the gap between classical and robust optimization may have been more appreciable.

Another issue with both Markowitz formulations was turnover, as they averaged a quarterly turnover above 46% compared to the 2.44% associated with the equal-weight strategy. In the real world, high turnover creates transaction costs and taxes, which would make the strategy far less attractive.

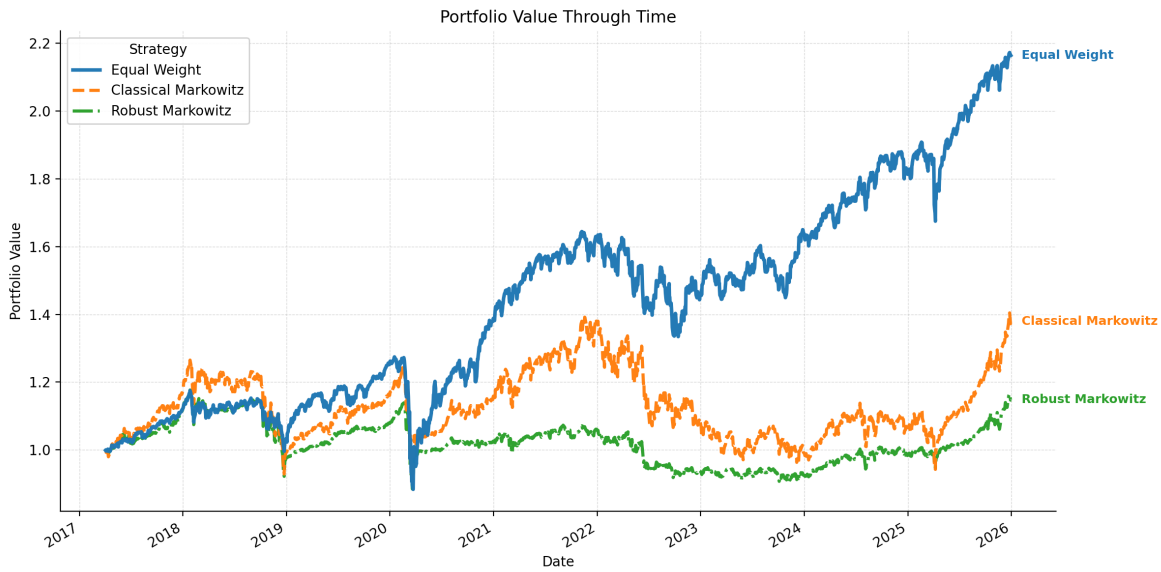


Fig. 1. Portfolio value over time.

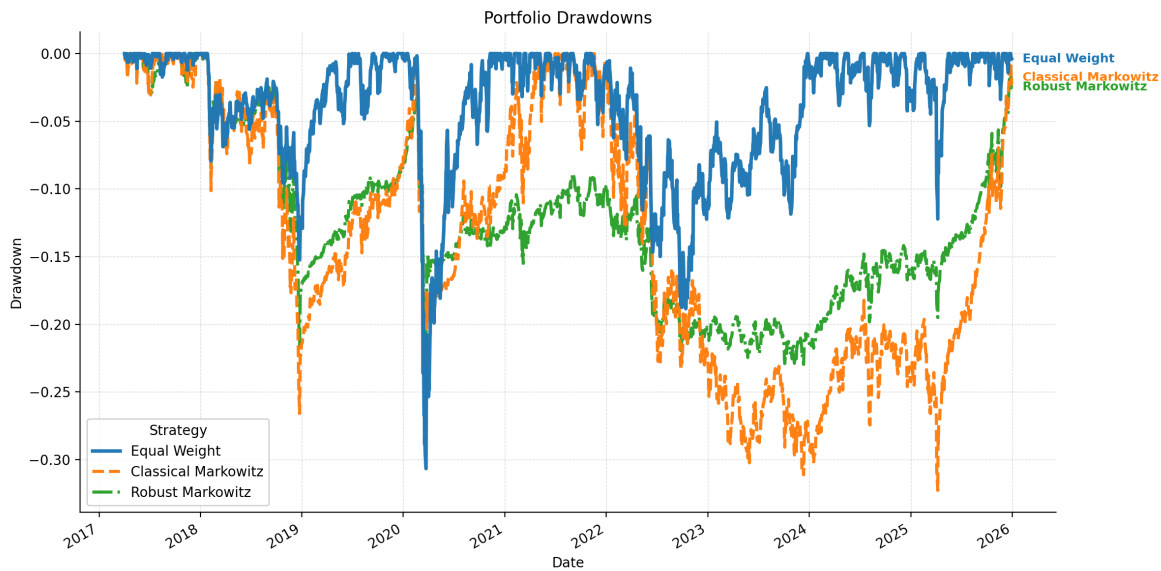


Fig. 2. Portfolio drawdowns.

TABLE I
PERFORMANCE METRICS

Strategy	Cum. Return	Ann. Return	Ann. Vol.	Sharpe	Max DD	Turnover	Final Value
Equal Weight	1.1637	0.0925	0.1402	0.6594	-0.3066	0.0244	2.1637
Classical Markowitz	0.3795	0.0376	0.1650	0.2276	-0.3227	0.4612	1.3795
Robust Markowitz	0.1478	0.0159	0.0979	0.1626	-0.2480	0.5190	1.1478

Our project presents several potential areas of further development. First, the parameter selection process for the Markowitz frameworks should be made more systematic. Next, the frameworks should explicitly factor in transaction costs so that these costs are not left at the end as a lingering drawback that we cannot quantify. Lastly, it is worth returning to the

original clustering based universe selection step laid out in our original proposal. The issue of high turnover rates can be especially prevalent when our ETF universe contains funds with similar attributes. Small changes in estimated returns and covariances can often cause the optimizer to switch between these ETFs, which leads to unnecessary turnover.

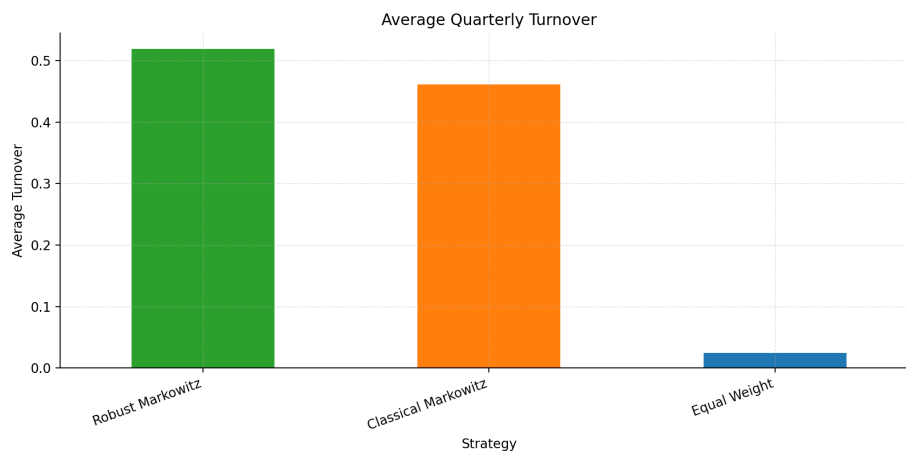


Fig. 3. Average quarterly turnover.

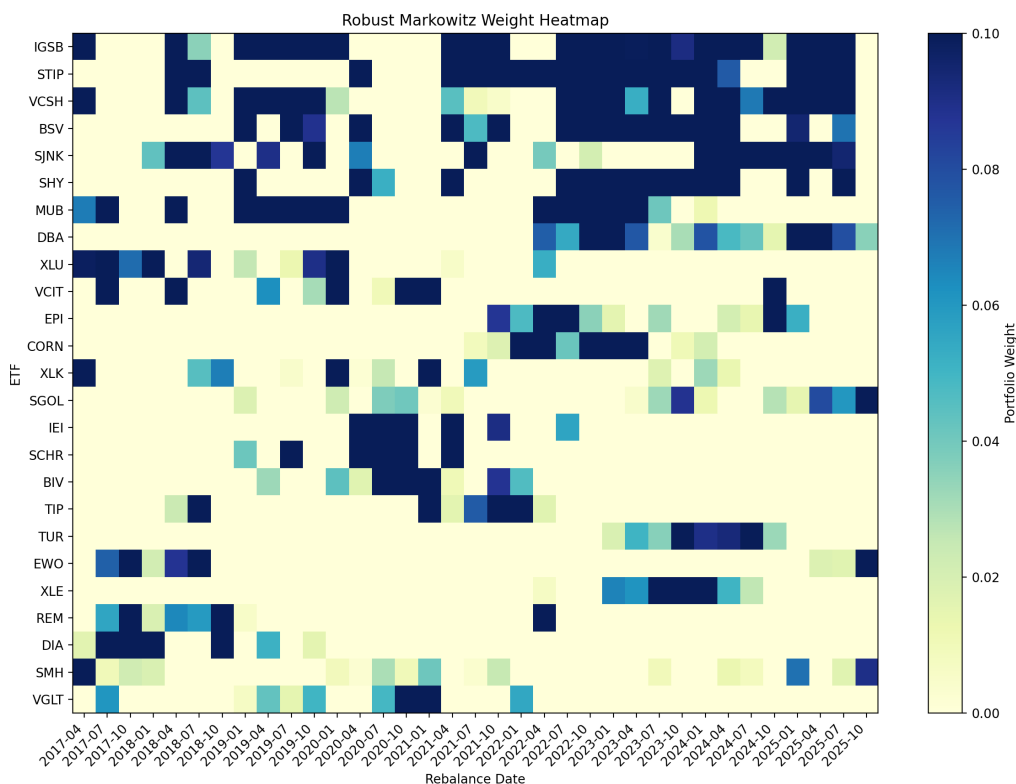


Fig. 4. Robust Markowitz weight heatmap for the 25 ETFs with the largest average weights.

VI. CONCLUSION

This project implemented a full Python backtesting pipeline for robust portfolio optimization in a broad ETF universe. The system downloads and cleans historical ETF data, computes daily returns, estimates return and covariance inputs using a rolling training window, and then compares three quarterly re-balanced strategies, namely equal weight, classical Markowitz, and robust Markowitz.

The results were clear. Equal weight performed best overall, with a cumulative return of 1.1637, annualized return of 9.25%, Sharpe ratio of 0.6594, and final portfolio value of

2.1637. Classical Markowitz and robust Markowitz produced lower returns and lower Sharpe ratios, while also generating much higher turnover. Robust Markowitz reduced volatility and maximum drawdown relative to classical Markowitz and also had a smaller drawdown than equal weight, but the improvement was not enough to offset its lower return.

More generally, there is the takeaway that portfolio optimization is highly sensitive to estimation quality and implementation choices. Robust optimization can make a portfolio more stable, but it does not guarantee better performance. In this project, simple diversification beat both optimized strategies, because it avoided overreacting to noisy rolling estimates and

kept turnover low. This finding supports one of the main motivations behind robust portfolio research: optimization must be used carefully, especially when the inputs have an appreciable level of uncertainty. It also motivates future work on clustering based universe selection, which could potentially reduce redundancy and help the optimizer make more stable decisions.

GROUP CONTRIBUTIONS

Andrew Gu helped establish the core objectives and boundaries of the project. He connected the idea to the relevant finance and portfolio construction literature, wrote the literature review, and interpreted the experimental results from the code.

Derrick Chu worked heavily on the mathematical formulation of the project. He worked on construction of the ETF universe, data processing, and the overall preparation for the backtesting pipeline.

Kovid Capildeo focused on implementing the optimization techniques and backtesting. He helped build the quarterly rebalancing process, debugged the optimizer and pipeline, and generated the final plots and tables with Matplotlib. He also did research on the Markowitz model to help write the background and related work section.

All group members assisted with debugging, reviewing the final results, and preparing the final report.

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